PRODUCT CONCEPT / ACADEMIC PROJECT

ExportPilot AI

An AI-assisted export document and compliance review platform for Indian exporters.

One platform to prepare shipment records, spot inconsistencies, review source-backed requirements, and help a human expert decide what needs attention.

WHO IT HELPS

Small and medium-sized Indian exporters and shipment documentation teams.

WHAT IT CHANGES

A scattered, largely manual checking process becomes one structured review workflow.

CORE VALUE

Find missing documents and mismatched fields before filing or shipment.

IMPORTANT BOUNDARY

Provides decision support, not guaranteed compliance or automated customs approval.

Scope note

This document explains the product idea and intended direction. Current capabilities are summarized from the public repository README; the proposed roadmap and success measures are initial hypotheses.

01 / OPPORTUNITY

A costly task spread across too many tools

Export documentation requires numerous forms and consistent shipment details. A simple mismatch can trigger extra review, correction, or follow-up. Smaller teams often coordinate this work through spreadsheets, shared folders, email, external websites, and customs advisers.

Who is the product for?

Primary user

A documentation executive or export operations manager preparing and reviewing shipment files.

Economic buyer

An owner or operations head at a small or medium-sized Indian export business.

Core user problems

- **Missing visibility:** Teams may not know which documents are required for a specific shipment.
- **Inconsistent details:** Values, weights, buyer information, HS/HSN codes, ports, and Incoterms may disagree across files.
- **Fragmented information:** Compliance requirements and potential incentives need source checking, not guesswork.
- **Late discovery:** Problems may be found only when a broker, bank, or customs reviewer checks the file.

Job to be done

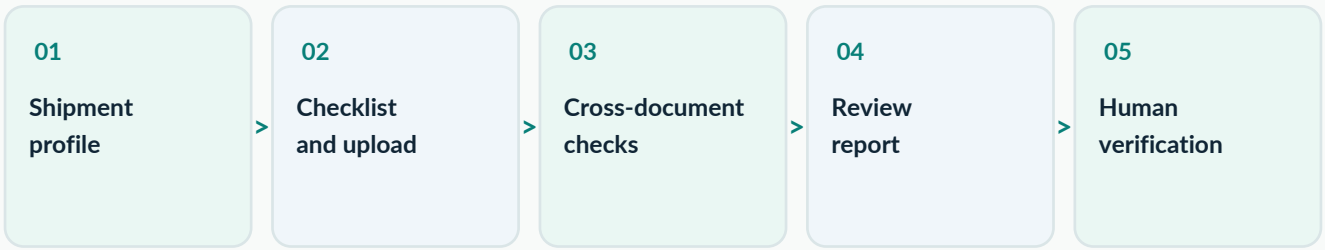
> **"Before I send a shipment file for processing, help me check that the right documents are present and their key details are consistent - and show me what still needs an expert to confirm."**

Starting assumption: Indian SME exporters will value a focused pre-submission review tool, even when they already use spreadsheets, accounting software, or a customs broker. This should be validated with user interviews.

02 / WORKFLOW

A structured path from shipment to review

The proposed experience follows five steps. The system guides the user through preparing a shipment and produces a review report instead of silently changing or submitting official records.



Walkthrough: shipping textiles from India to Canada

- **Create a shipment:** Record the product category, HS/HSN code, destination, trade terms, shipment method, and value.
- **Get a checklist:** See documents that may be needed, with uncertain or context-specific requirements flagged.
- **Upload documents:** Attach the commercial invoice, packing list, transport document, and other records.
- **Check consistency:** Compare selected fields; for example, if the invoice lists 850 cartons but the packing list shows 805, show a discrepancy.
- **Review and resolve:** Present findings with action items and supporting evidence for the exporter or customs professional.

Output: a shipment-specific checklist, a discrepancy report, a prioritized action list, and clearly marked questions for human verification.

What the product does not do

It does not guarantee that a shipment is legally compliant, calculate unsupported incentive rates, or automatically file documents to government portals. Final compliance and filing decisions stay with responsible people.

03 / PRODUCT FEATURES

What ExportPilot AI contains

Module	Purpose / value
Shipment profile	Collect key details: product, HS/HSN code, destination, Incoterm, payment terms, shipment mode, and value.
Document checklist	Generate a shipment-specific list of documents to prepare and review.
File management	Upload and label files, with organization-specific access boundaries.
Document checks	Flag differences in names, quantities, weights, codes, ports, and other matched fields.
Verification report	Summarize findings, missing inputs, unresolved questions, and recommended actions.
Incentive support	Use reviewed, versioned rate records and show when rate information is missing.
Compliance checker	Use approved, source-backed requirement records with dates and confidence indicators.
Expert handoff	Support review by a CHA, customs broker, or DGFT consultant rather than replacing them.

Responsible use of AI

- Use AI to assist with checklists and explanations, while using repeatable cross-document rules where possible.
- Show the source, effective date, confidence, and unresolved items for any relevant compliance information.
- Never fabricate government incentive rates or fill gaps with unsupported claims.
- Ask for human review on ambiguous, outdated, or high-impact regulatory questions.

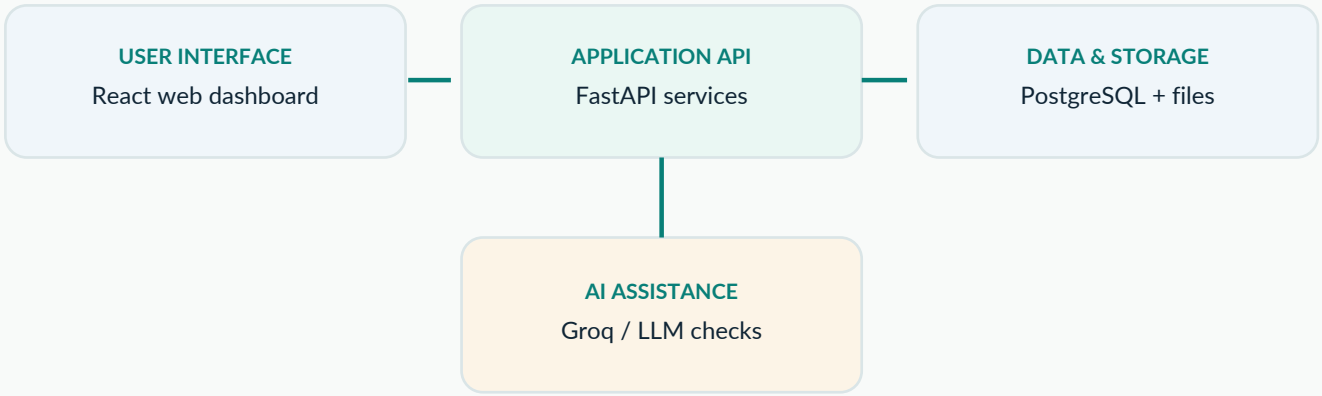
Design principle: useful automation + visible evidence + a human decision point.

04 / TECHNICAL APPROACH

How the system could work

The GitHub README describes an existing multi-tenant web application with a FastAPI backend, PostgreSQL database, React frontend, and Groq-based AI services. The diagram below is a simplified conceptual view.

Illustrative logical architecture based on the repository README



Core data and services

Layer	Responsibilities
React web interface	Shipment screens, uploads, dashboards, checklists, and review results.
FastAPI backend	Authentication, shipment workflows, validation, document processing, and API endpoints.
PostgreSQL + file storage	Organizations, shipment records, documents, rate data, approved sources, and audit history.
AI assistance	Suggested checklists, guided explanations, and supported interpretation of detected issues.

Security and reliability needs

- Separate each exporter organization’s data; grant access through role-based permissions.
- Log changes to important records, protect uploads, and avoid leaking files across tenants.
- Retain provenance for reviewed compliance sources and versioned rates.
- Test document comparison, negative cases, missing-source scenarios, and access controls.

Implementation note: The repository README lists these technologies and multiple Phase 1 endpoints. This brief does not independently verify production readiness or external regulatory accuracy.

05 / POSITIONING

Why a user would choose it

The primary alternative is manual checking across spreadsheets, email, cloud folders, official websites, and conversations with a customs broker. ExportPilot AI aims to make the preparation and first-pass audit more organized.

Approach	Works well for	Potential limitation
Spreadsheets + email	Flexible, familiar, inexpensive to start	Repeated data entry; comparison and status tracking depend on individuals
Customs broker / consultant	Professional context and expert judgment	Often becomes the final review stage rather than an early self-service check
General ERP / logistics tool	Broad shipment and business workflows	May not focus on cross-document error explanation for small exporters
ExportPilot AI	Shipment-focused checks, evidence, and review handoff	Not a substitute for a licensed professional; needs reliable sources

Meaningful differentiators

- **Workflow-specific:** The process centers on checking an export shipment, not generic file storage.
- **Cross-document view:** Highlights inconsistencies between related shipment records.
- **Source-backed information:** Displays available evidence and refuses to invent missing rate data.
- **Human review:** Makes the remaining decision and verification work visible.

What still needs proof

The comparison above is a positioning hypothesis, not an independently tested claim against named commercial competitors. Customer interviews and competitor trials should test whether these differences save meaningful time or prevent errors.

06 / SCOPE AND DELIVERY

Build the smallest useful workflow first

Initial MVP

- Create and manage shipment profiles.
- Generate a tailored document checklist and upload basic shipment files.
- Compare a small set of high-impact fields across documents.
- Produce a clear discrepancy list with human-review status.
- Keep an audit trail and basic organization-level access controls.

Next increments

Stage	Possible increment	Reason
After MVP	More reliable field extraction and stronger validation test coverage	Reduce missed and false discrepancy flags
After MVP	Reviewed, versioned incentive data and richer source-backed compliance support	Expand decision support without guessing
Later	Expert handoff workflow, team collaboration, notifications, and reporting	Support end-to-end resolution across staff and advisers
Only if justified	Third-party government integrations with explicit user confirmation	Avoid fragile or unsafe automation of official submissions

How to judge whether it helps

- **Usability:** Time to prepare and complete the first review of one shipment.
- **Quality:** Proportion of known test discrepancies correctly surfaced, and false-positive frequency.
- **Adoption:** Number of shipments reviewed and repeat use by pilot teams.
- **Trust:** Percentage of findings linked to reviewable documents or sources.

These are proposed measures, not verified results or promised improvements.

07 / VISION & NEXT STEPS

The product vision in six statements

Element	Vision statement
FOR	Small and medium-sized Indian exporters and their shipment documentation teams.
WHO	Need to prepare and check multiple export documents, catch inconsistencies, and find reliable information on applicable requirements.
THE	ExportPilot AI is a web-based, AI-assisted export document verification platform.
THAT	Helps teams spot missing information and mismatched details before shipment or filing.
UNLIKE	Manually checking documents through spreadsheets, email, and scattered websites.
OUR PRODUCT	Unifies shipment documents, consistency checks, and source-backed compliance and incentive information with transparent human-review points.

The main risks to validate

- **Incorrect advice:** Regulatory requirements can vary and change; require reviewed sources and prominent caveats.
- **Bad input quality:** Scans and uploaded documents may be incomplete or inconsistent.
- **Weak adoption:** Small exporters may prefer existing methods unless benefits are tangible.
- **Data sensitivity:** Shipment and customer documents require careful access control and retention policies.

What to place in the course repository

Required: product-vision.md with the six assignment elements. **Optional:** This concept PDF as additional project documentation. The Moodle submission should be the public GitHub link to product-vision.md, not merely the repository homepage.

Repository: github.com/rvashi19/kalypto
Product concept based on the repository README reviewed on 27 October 2026.

This document is a project concept and learning artifact, not customs, financial, or legal advice.